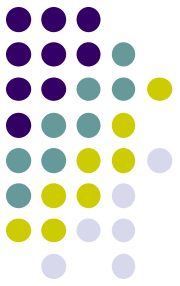


Regression Analysis

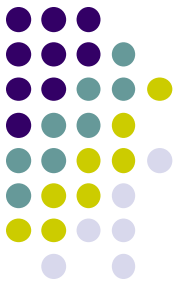
Method of **Regression Analysis** is used to **forecast** or **estimate** values of one variable (*respond variable, predicted variable*) by certain formula of one or several other variables (*descriptive variables, estimators*)

Example. There certain relation between **height** and **weight** of student. Based on data collected from n students, how to estimate **weight** of another student if his **height** is given?



Simple Linear Regression

- Simple linear regression involves one independent variable and one dependent variable.
- The relationship between the two variables is approximated by a straight line.

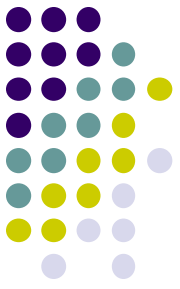


Simple Linear Regression Model:

$$Y = a.X + b + e$$

where

- * a is called *slope* of regression equation, informing how much the dependent variable Y grows up (or gets down) if the independent variable X increases 1 unit;
- * b is called *regression constant* (*intercept*), showing the *intersection* point of regression line and vertical axis, that is the value of Y when X takes value 0 ;
- * e is *residual* of regression, indicates error of estimation at each point of observation.

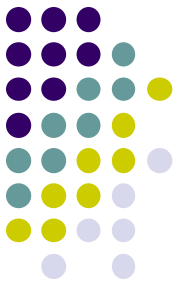


Example. Relation between “Expenditure for buying tems” (furniture, TV, motorbike, etc.) and “Income from trading” of households: Linear regression equation with

“Income from trading” = independent variable X

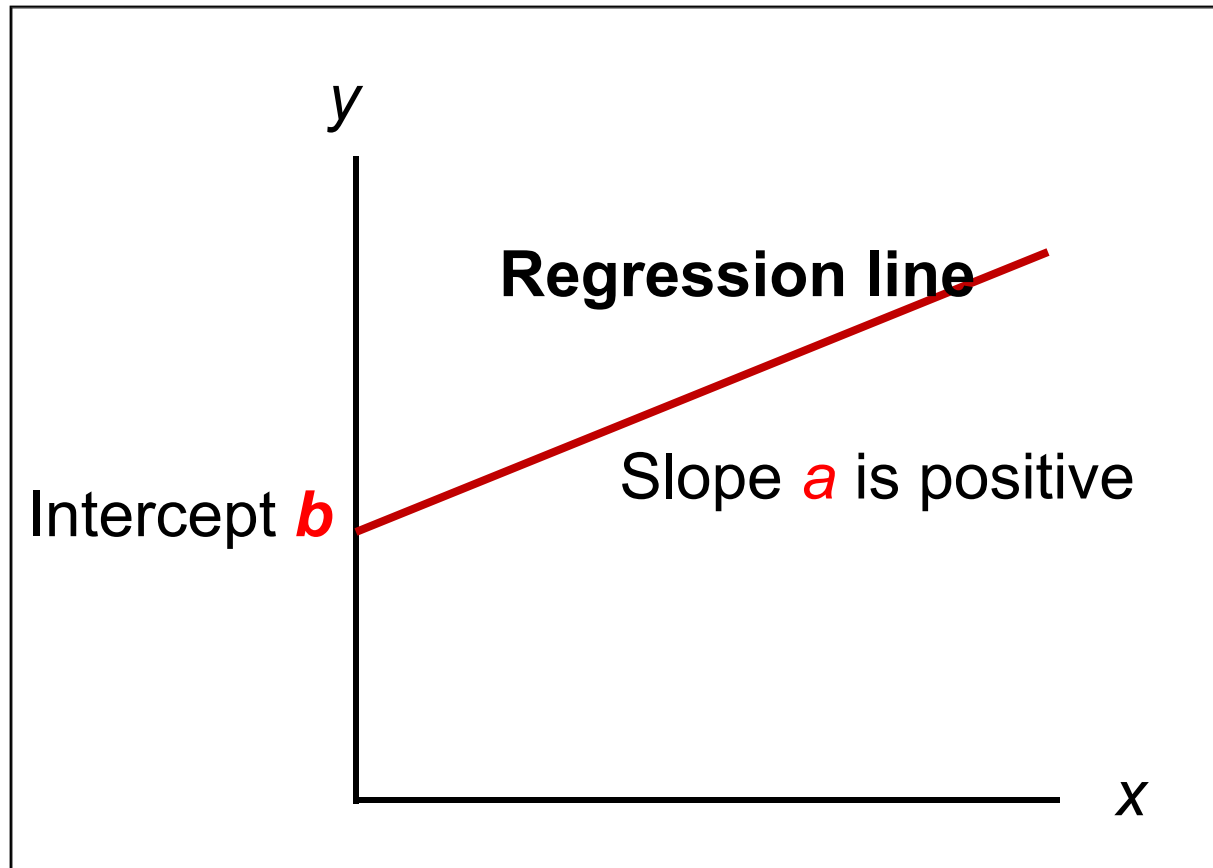
“Expenditure for buying valued items” = dependent variable Y .

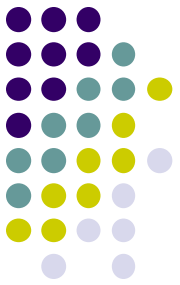
- Slope a = the share for “buying valued items” in 1 million VND of “income from trading”
- Intercept b = expenditure for buying valued items of given household when the household has no income from trading



Simple Linear Regression Equation

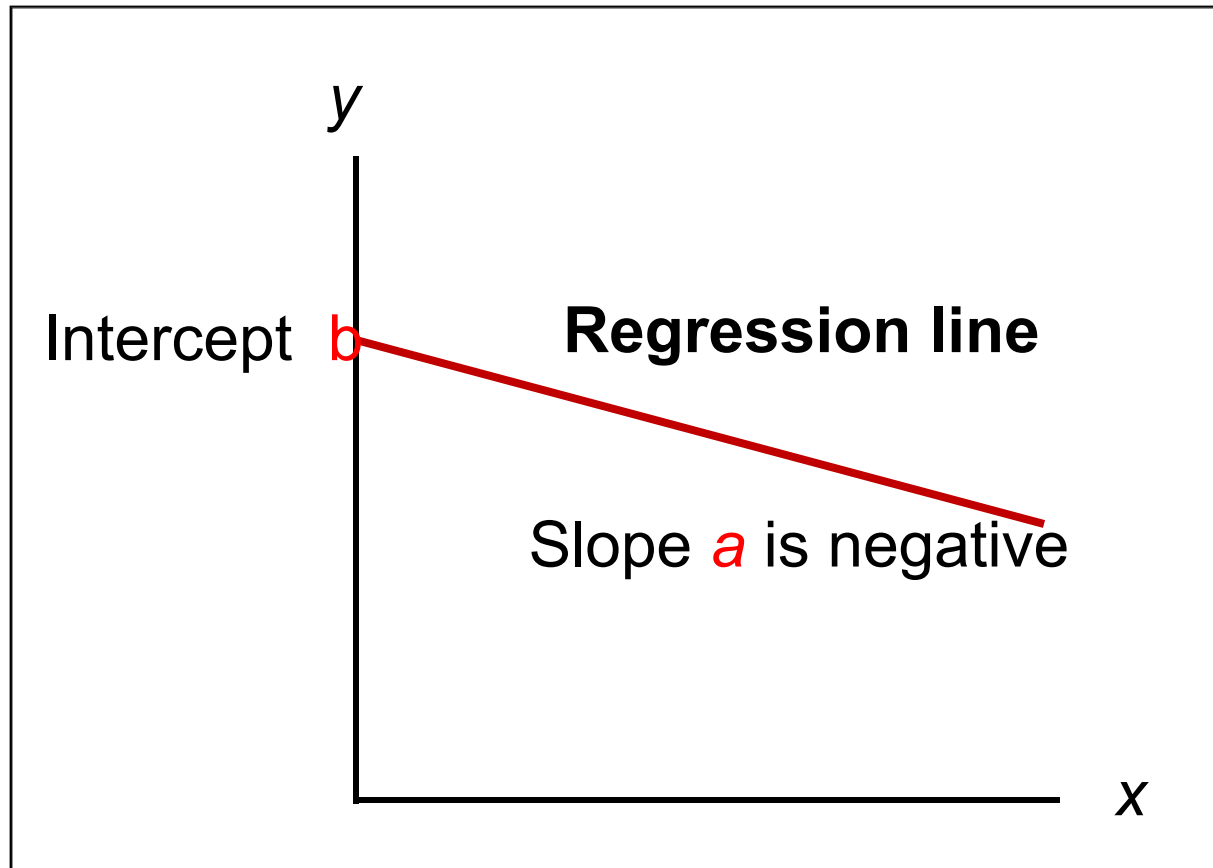
- Positive Linear Relationship

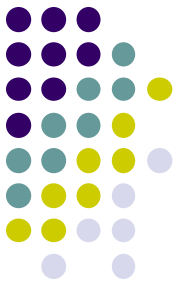




Simple Linear Regression Equation

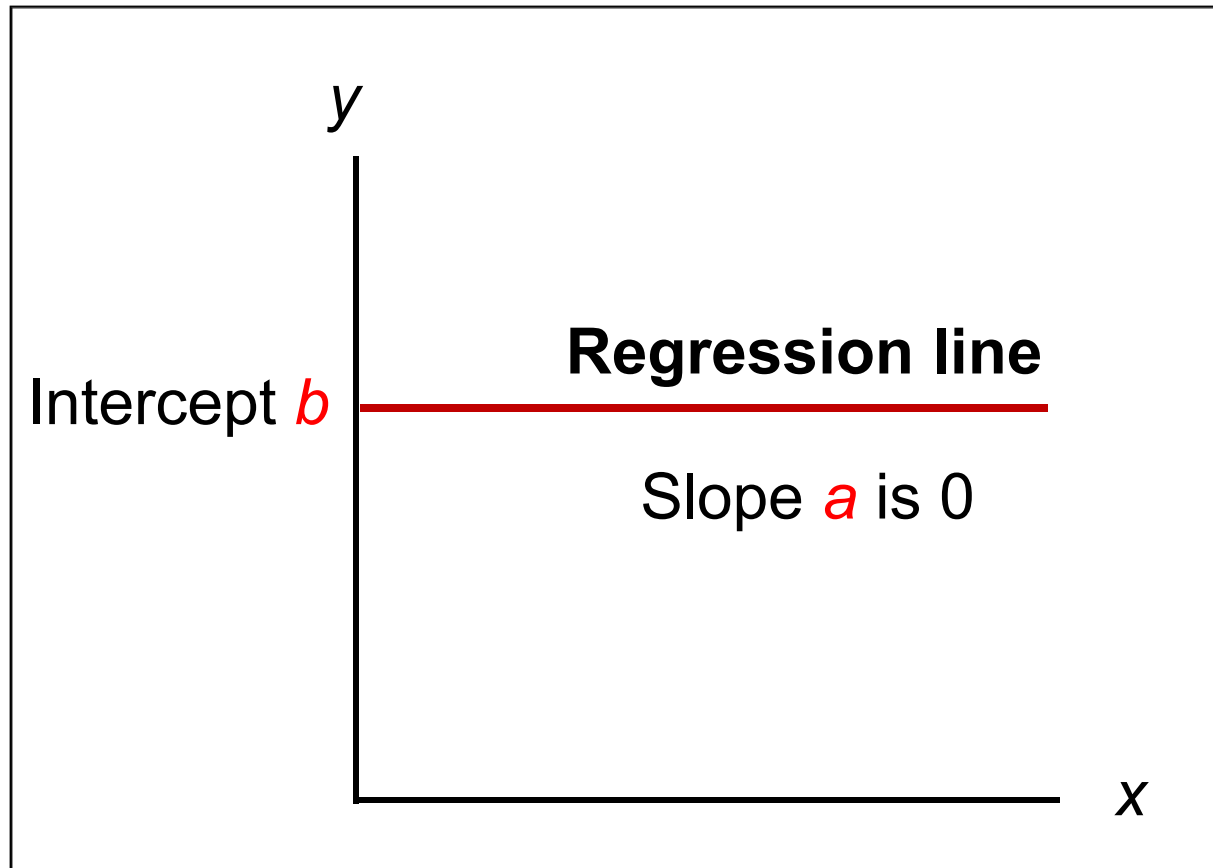
- Negative Linear Relationship



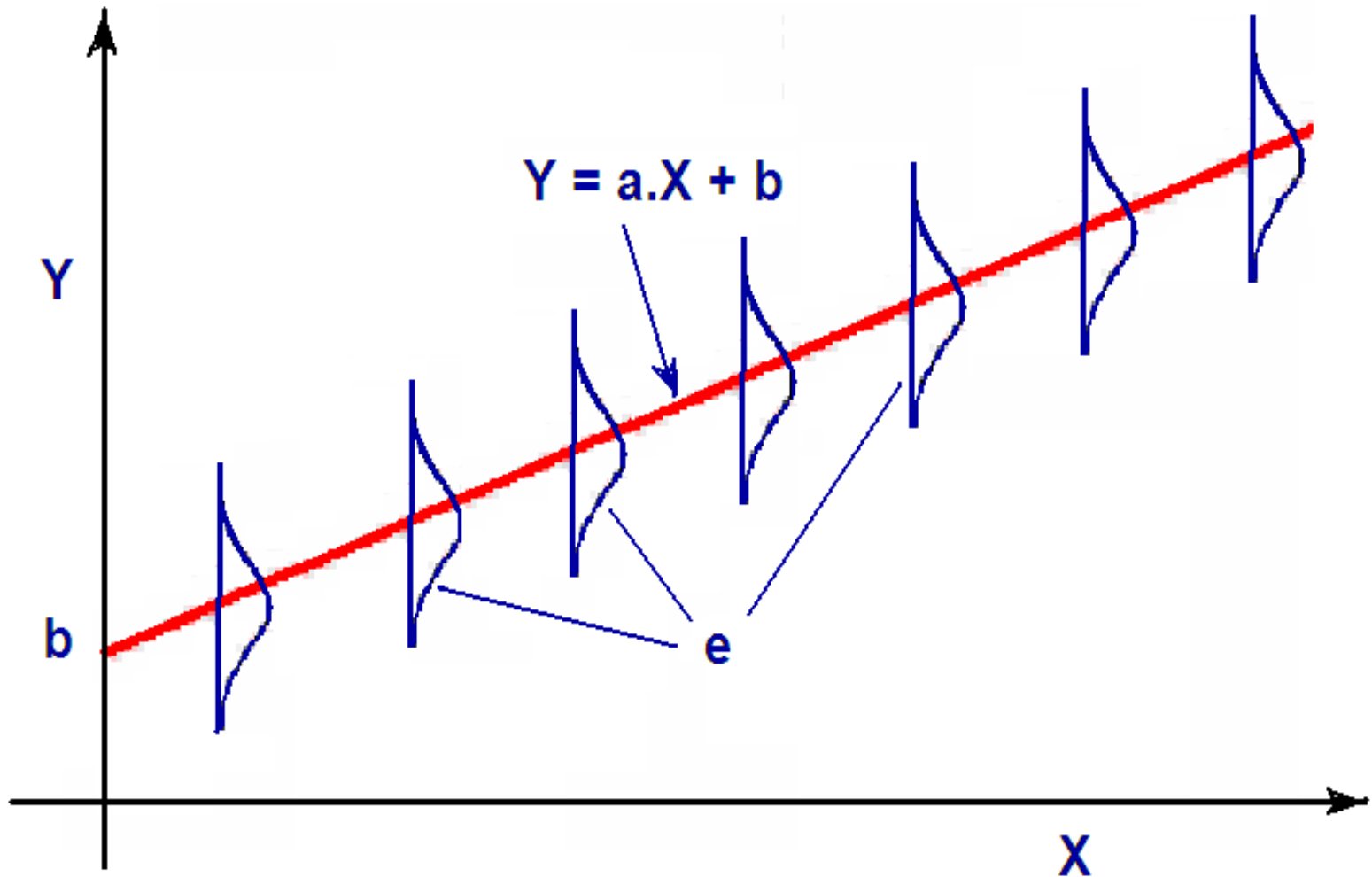
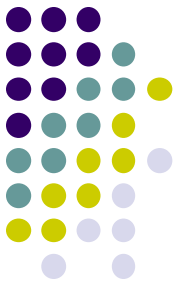


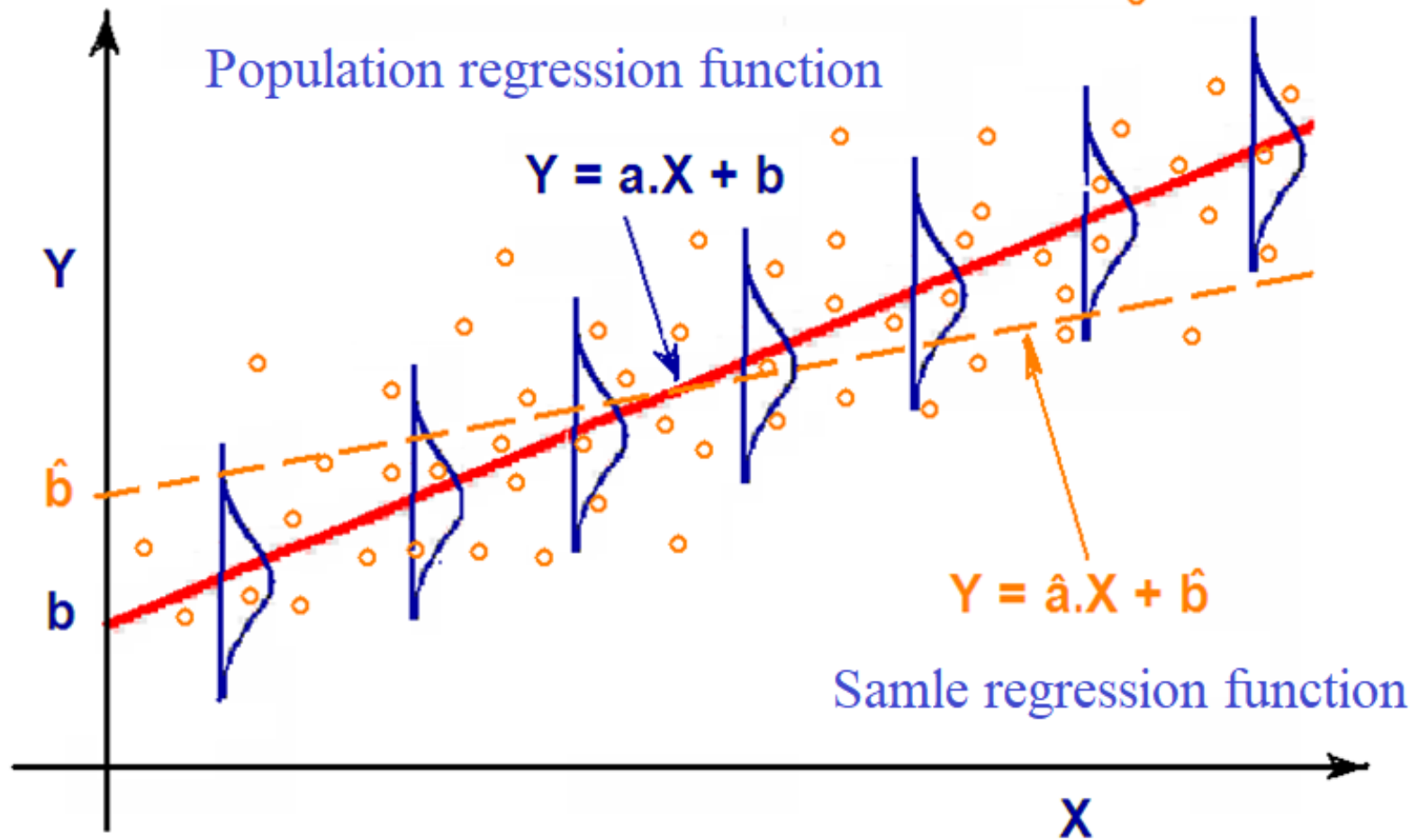
Simple Linear Regression Equation

- No Relationship

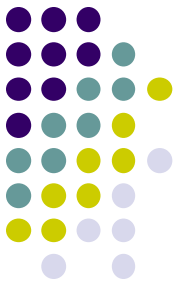


Linear Regression model





$$Y_i = \hat{Y}_i + \hat{e}_i = \hat{a}.X_i + \hat{b} + \hat{e}_i$$



Estimate regression coefficients using method of least squares criterion

For linear regression model $Y = a.X + b + e$, collect a sample

$$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$$

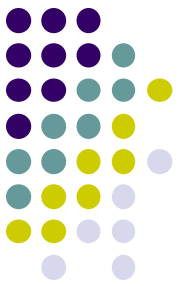
Regression function should be of the form

$$\hat{Y}_i = \hat{a}.X_i + \hat{b}$$

$$Y_i = \hat{Y}_i + \hat{e}_i = \hat{a}.X_i + \hat{b} + \hat{e}_i$$

→ Need to estimate the regression coefficients minimizing the sum of residual (error) squares:

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{a}X_i - \hat{b})^2 = f(\hat{a}, \hat{b}) \rightarrow \min$$



Solution

Partial derivatives of function f vanish at the minimal point of f (sufficient condition):

$$\frac{\partial f}{\partial \hat{b}} = -\sum_{i=1}^n 2(Y_i - \hat{a} \cdot X_i - \hat{b}) = 0$$

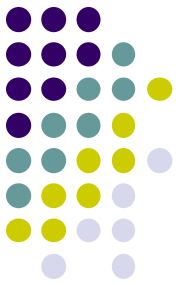
$$\frac{\partial f}{\partial \hat{a}} = -\sum_{i=1}^n 2(Y_i - \hat{a} \cdot X_i - \hat{b})X_i = 0$$

Then

$$\sum_{i=1}^n Y_i = \hat{a} \sum_{i=1}^n X_i + n\hat{b} ; \sum_{i=1}^n Y_i X_i = \hat{a} \sum_{i=1}^n X_i^2 + \hat{b} \sum_{i=1}^n X_i$$

$$\hat{a} = \frac{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^n X_i^2 - n\bar{X}^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} ; \hat{b} = \bar{Y} - \hat{a}\bar{X}$$

2. Evaluation of model quality



Having **estimated regression coefficients**, perform regression function

$$\hat{Y}_i = \hat{a}.X_i + \hat{b}$$

$$Y_i = \hat{Y}_i + \hat{e}_i = \hat{a}.X_i + \hat{b} + \hat{e}_i$$

→ For each value of independent in the regression equation we have value of a new variable $\hat{Y}$. This is **prediction** of dependent variable Y .

* Correlation coefficient $R = r(Y, \hat{Y})$ between dependent variable Y and prediction variable $\hat{Y}$ represents the “closeness” between dependent and prediction variable.

* $0 < R = r(Y, \hat{Y}) < 1$

* $R = r(Y, \hat{Y}) = |r(Y, X)|$

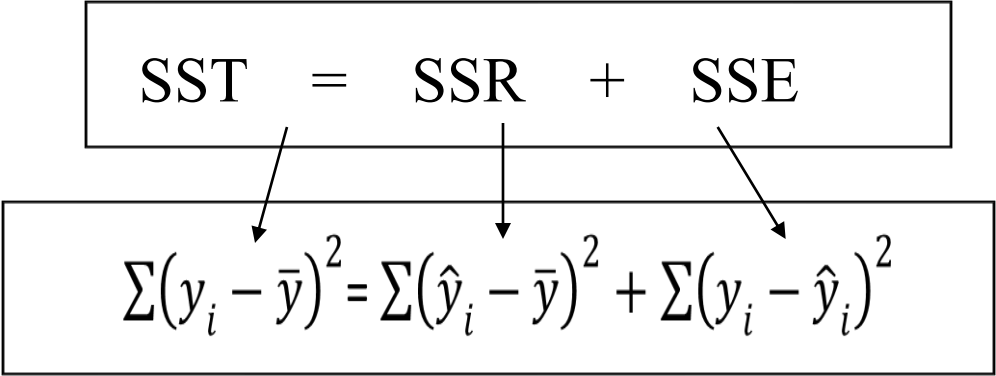
* For simple linear regression model, R equals absolute value of correlation coefficient between dependent and independent variables $|r(X, Y)|$

* In practice, the quantity R^2 is usually used in place of R . This quantity is called **coefficient of determination**, reflects the percentage variability of the dependent variable contained in the regression equation.

* The model is of good quality if $R^2 > 60\%$,
of poor quality if $R^2 < 30\%$,
still usable if R^2 is between 30% and 60%

Coefficient of Determination (Alternative approach)

$$\text{SST} = \text{SSR} + \text{SSE}$$


$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

where:

SST = total sum of squares

SSR = sum of squares due to regression

SSE = sum of squares due to error

Coefficient of Determination

- The coefficient of determination is:

$$R^2 = SSR/SST$$

where:

SSR = sum of squares due to regression

SST = total sum of squares

Testing for Significance

- To test for a significant regression relationship, we must conduct a hypothesis test to determine whether the value of α is zero.
- Two tests are commonly used:

t Test

and

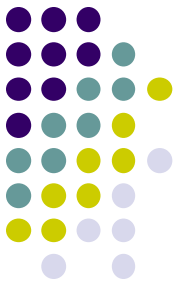
F Test

- Both the t test and F test require an estimate of σ^2 , the variance of ε in the regression model.

- Theorem. Assuming that the errors of the regression model are independent and have Normal distributions with zero mean value and the common variance, the regression parameters are estimated the ordinary least squares (OLS) method. Then the two statistics

$$\frac{\hat{a}}{se(\hat{a})} \quad ; \quad \frac{\hat{b}}{se(\hat{b})}$$

have Student distribution with $(n-2)$ degrees of freedom.



- Variation (variance) of regression error

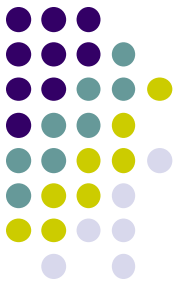
$$\hat{\sigma}^2(e) = \sum_{i=1}^n \hat{e}_i^2 / (n-2) = \sum_{i=1}^n (\hat{Y}_i - Y_i)^2 / (n-2)$$

- Variation (variance) of slope estimation

$$Var(\hat{a}) = \frac{\hat{\sigma}^2(e)}{\sum_{i=1}^n (X_i - Mean(X))^2}$$

- Standard Error of slope estimation

$$se(\hat{a}) = \sqrt{Var(\hat{a})}$$



- Variation (variance) of intercept estimation

$$Var(\hat{b}) = \frac{\hat{\sigma}^2(e)}{\sum_{i=1}^n (X_i - Mean(X))^2} \cdot \frac{\sum_{i=1}^n X_i^2}{n}$$

- Standard Error of intercept estimation

$$se(\hat{b}) = \sqrt{Var(\hat{b})}$$

Test 1.

Null Hypothesis

$$H_0: a = 0$$

Alternative Hypothesis

$$H_a: a \neq 0$$

With a variable $T^{(n-2)}$ of Student distribution
with $(n-2)$ degrees of freedom

Use the statistic

$$t_a = \frac{\hat{a}}{se(\hat{a})}$$

to calculate probability (p-value)

$$p = P \{ |T^{(n-2)}| > |t_a| \}$$

- Compare the p-value to the significance level α .
 - If $p \geq \alpha \Rightarrow$ Accept the hypothesis, consider $a = 0$, between Y and X *there is no linear relationship*, increase of X *does not lead* Y to increase or decrease.
 - If $p < \alpha \Rightarrow$ Reject the hypothesis, assert $a \neq 0$, the conclusion between Y and X really has a strong linear relationship, the *increase* in the value of X will lead to an *increase* (if $a > 0$), or *reduce* (if $a < 0$) the value of Y .
- The test for intercept coefficient b is also carried out in the same way.

Confidence Interval for a and b

- The form of a confidence interval for a and b are:

$$\begin{aligned} & (a^{\wedge} - t_{1-\alpha/2} s_{a^{\wedge}}; a^{\wedge} + t_{1-\alpha/2} s_{a^{\wedge}}) \\ & (b^{\wedge} - t_{1-\alpha/2} s_{b^{\wedge}}; b^{\wedge} + t_{1-\alpha/2} s_{b^{\wedge}}) \end{aligned}$$

where

$a^{\wedge}$ and $b^{\wedge}$ are the point estimators,

$t_{1-\alpha/2} s_{a^{\wedge}}$ and $t_{1-\alpha/2} s_{b^{\wedge}}$ are the margins of error,

$t_{1-\alpha/2}$ is the t value providing an area of $\alpha/2$ in the upper tail of a t distribution with $n - 2$ degrees of freedom.

Confidence Interval for a (or b)

- We can use a confidence interval for a (or b) to test the hypotheses just used in the t test.
- H is rejected if 0 (the hypothesized value of a (or b)) is not included in the confidence interval for a (or b , respectively) .

Testing for Significance: F Test

- Hypotheses

$$H: a = 0$$

$$K: a \neq 0$$

- Test Statistic

$$F = \frac{R^2}{(1 - R^2)/(n - 2)}$$

Test 2.

Theorem. Consider the linear regression model

$$Y = a.X + b ,$$

with assumption of independent and Normal distributed residuals. Then the variable (statistic) F has Fisher distribution with $(1, n-2)$ degrees of freedom.

P-value Approach: Test the hypothesis H calculating the probability

$$p = P\{F^{(1, n-2)} > F\}$$

Then compare p with significance level α to decide accept or reject the hypothesis H .

* If $p \geq \alpha \rightarrow$ accept hypothesis H , conclude regression coefficients equal 0. Then independent variable has no influence on regression model, there is no association between that variable and dependent variable $\rightarrow$ The model is not correct, it need to find other models

* If $p < \alpha \rightarrow$ reject hypothesis H , confirm the slope differs from 0 and the model is good fitted

Using Excel to Compute Regression Parameters

- Excel has two functions for computing the parameters of Simple Linear Regression Model:
 - INTERCEPT is used to compute the intercept parameter of the model.
 - SLOPE is used to compute the slope parameter of the model.